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Sofia ML Regression 2026 Spring - A challenge to predict a target real value.

Executive Summary

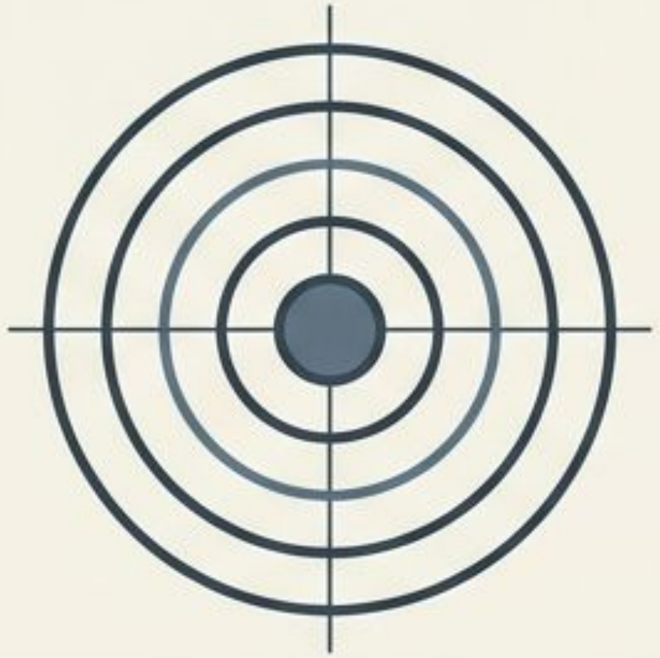
The Goal: Accurate Regression

The Sofia ML Regression 2026 Spring challenge is a supervised regression task. The objective is to predict a target real value from 15 anonymized multivariate features (x0 to x14).

Key Finding: Linear Models Fail

Initial analysis reveals that basic linear models achieve negative R^2 scores. This indicates the absence of strong, direct linear signals, necessitating the use of non-linear approaches.

The Rules of the Optimization Game

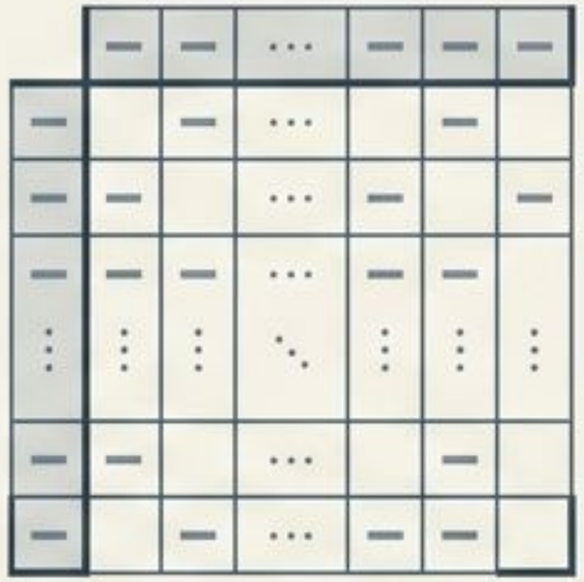


Supervised Regression

Task: Predict target probabilities perfectly based on the provided dataset.

Anonymized Inputs

15 multivariate features (x0 to x14) provided with no inherent semantic meaning.



$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

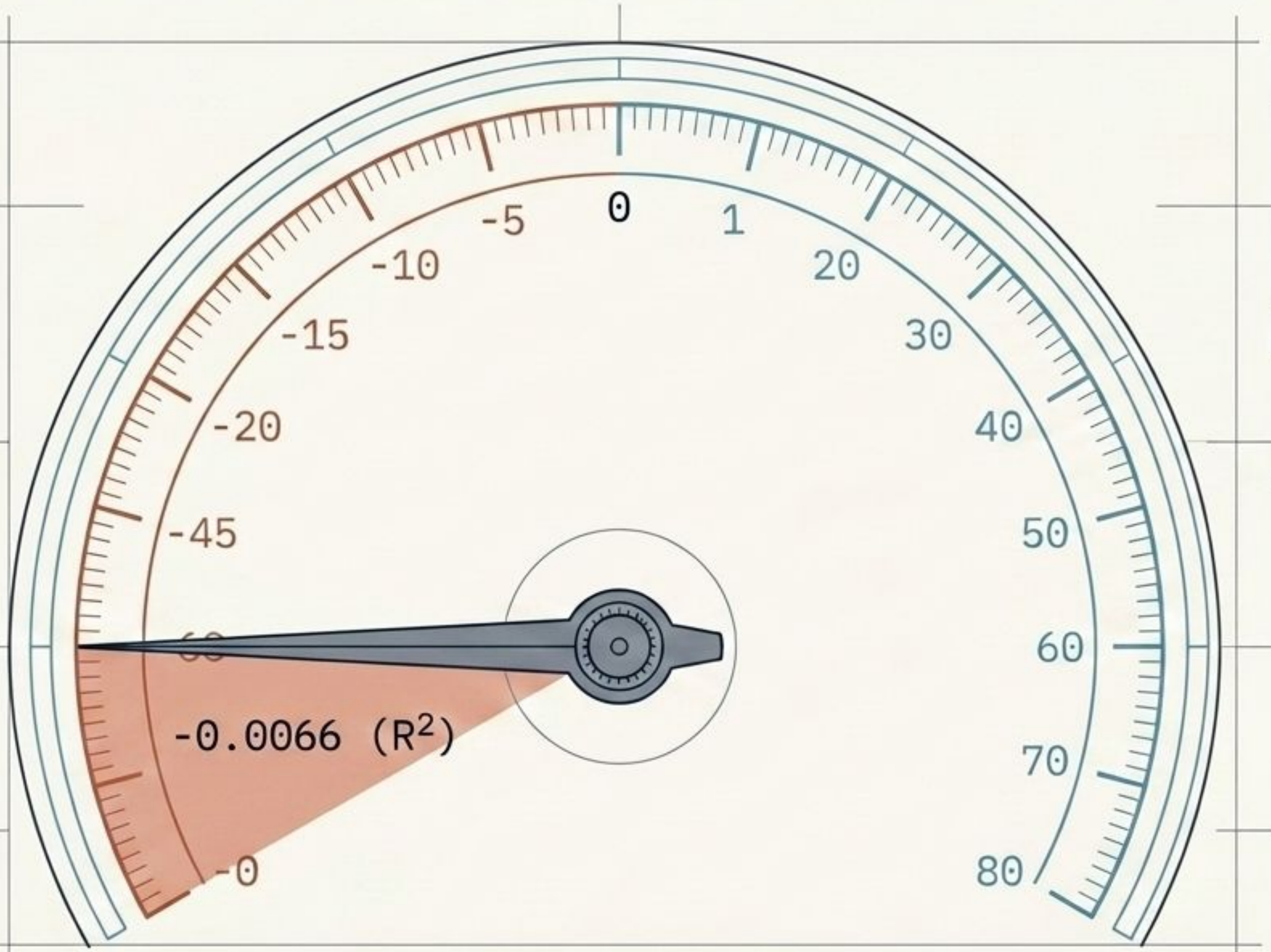
Coefficient of Determination

Evaluation Metric: Max value = 1 (achieved with a perfect, error-free prediction).

Avoid Public Overfit

Constraint: Initial ranking on the Public test does not guarantee Private test success. Generalization is paramount.

The Absence of Direct Linear Signals



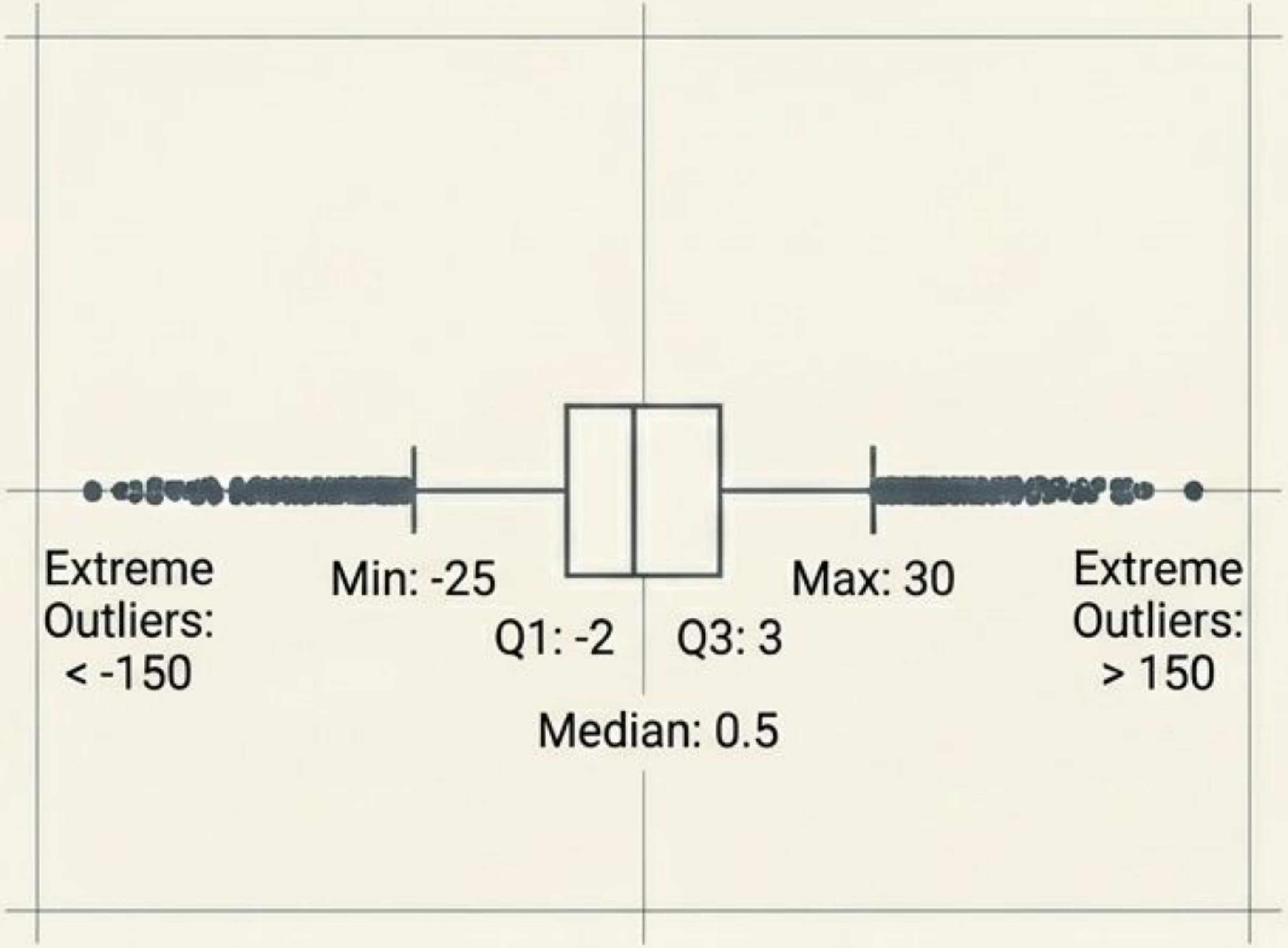
Linear_Regression	:	-0.0066
Ridge	:	-0.0042
Lasso	:	-0.0021
ElasticNet	:	-0.0035

Insight:

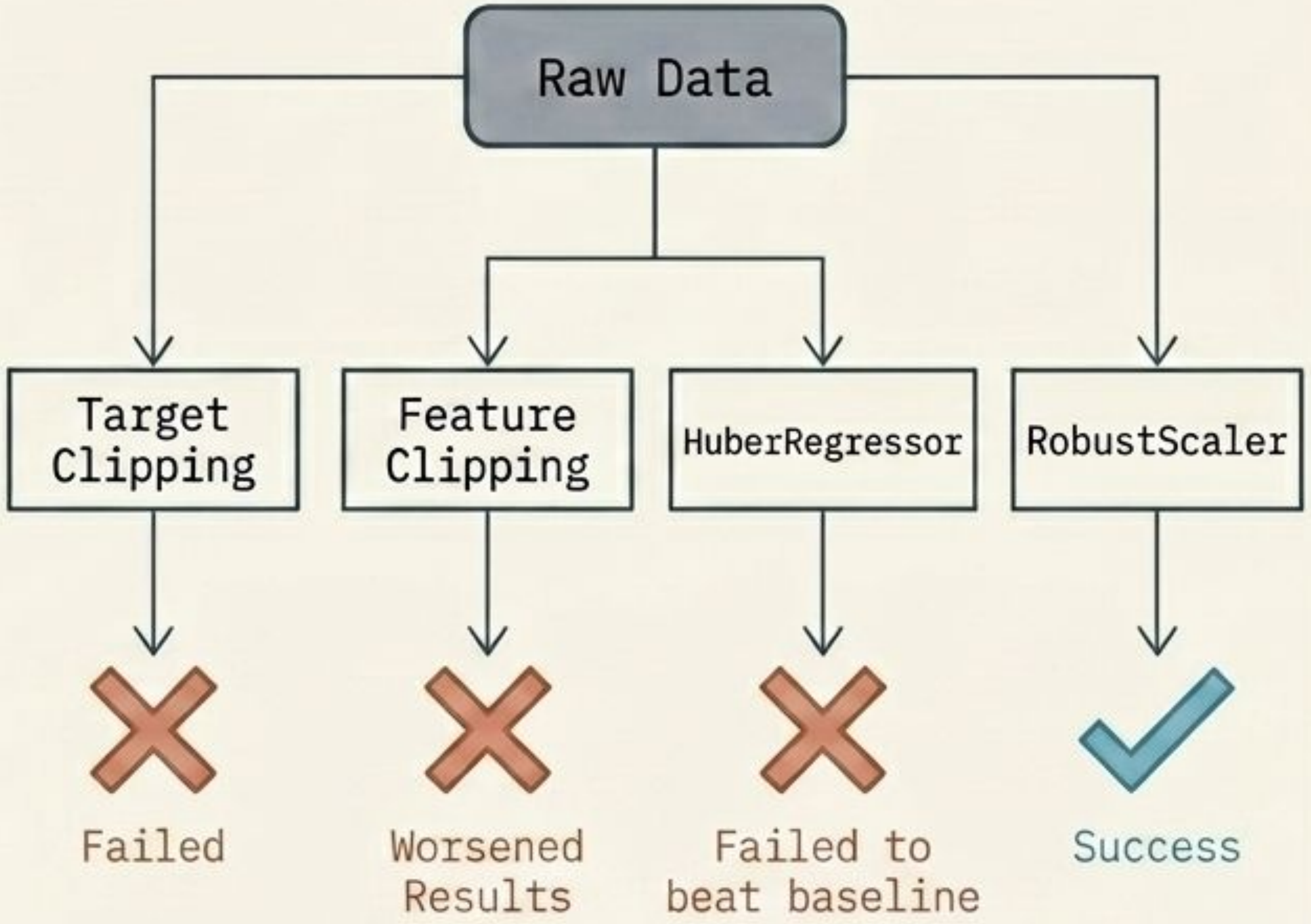
The R^2 metric heavily penalizes squared prediction errors. Models optimizing this require strict validation.

The original x0–x14 features inherently lack strong, direct linear predictability, instantly rendering basic linear models ineffective.

Handling the Heavy Tails

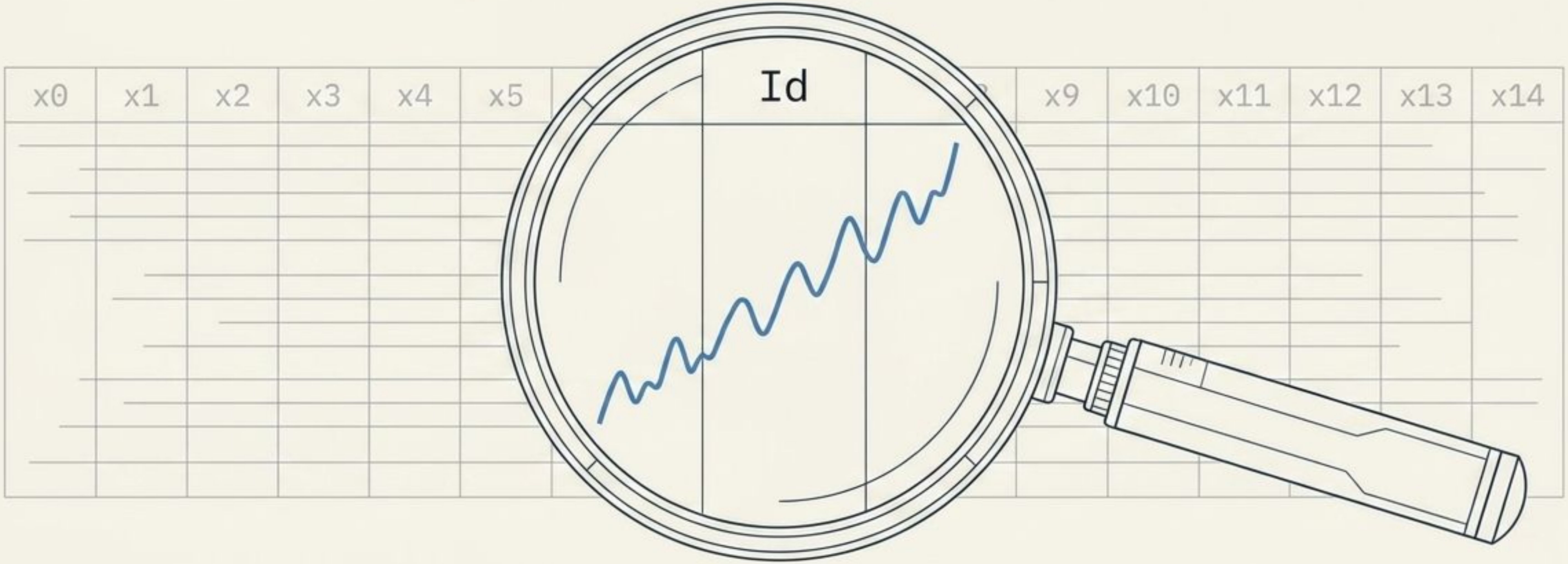


Features and targets exhibited extreme heavy outlier tails.



Core Insight: Clipping destroyed valuable information. Outliers contained critical signal. RobustScaler effectively managed extreme values while preserving the underlying data integrity.

The Id Anomaly



RobustScaler + x0...x14 + Id + Ridge = 0.0209 (Local CV)

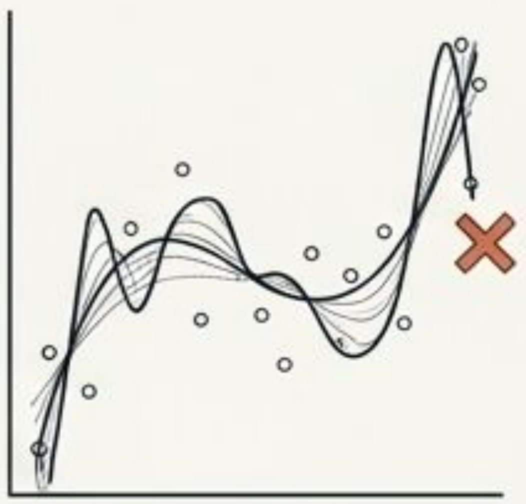
Typically dropped as meaningless, the Id column contained hidden distribution shifts and chronological ordering. Injecting it alongside the scaled features was the first major step into positive scoring territory.

The Model Graveyard

Over-Complex & Overfit

- Poly Degree 2 + Ridge
- Poly Degree 3 (Weak Reg)

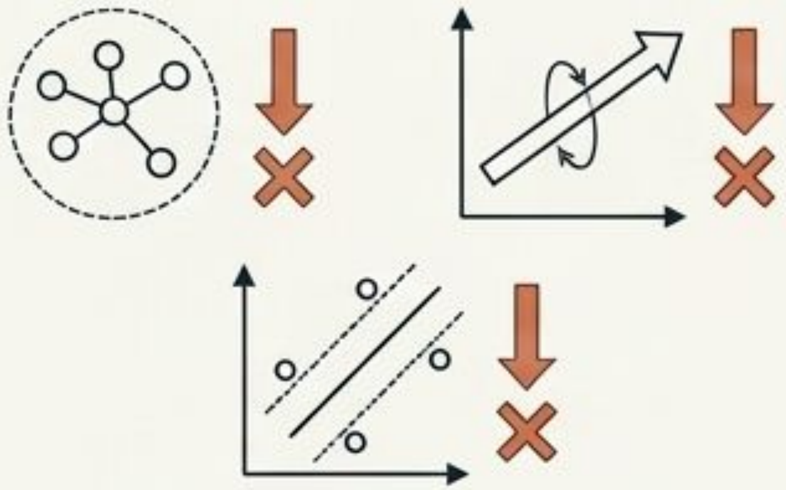
Note: Degree 3 heavily regularized reached 0.0089, but expansions quickly overfit the data.



Under-Performers

- KNN
- PLS Regression
- SVR

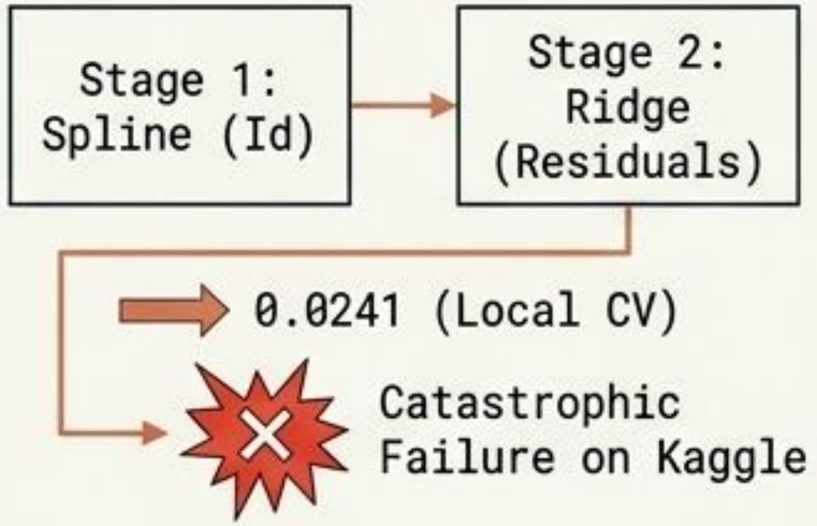
Note: Failed to capture useful signal. PLS models plummeted back to negative baseline levels.



Misguided Structure

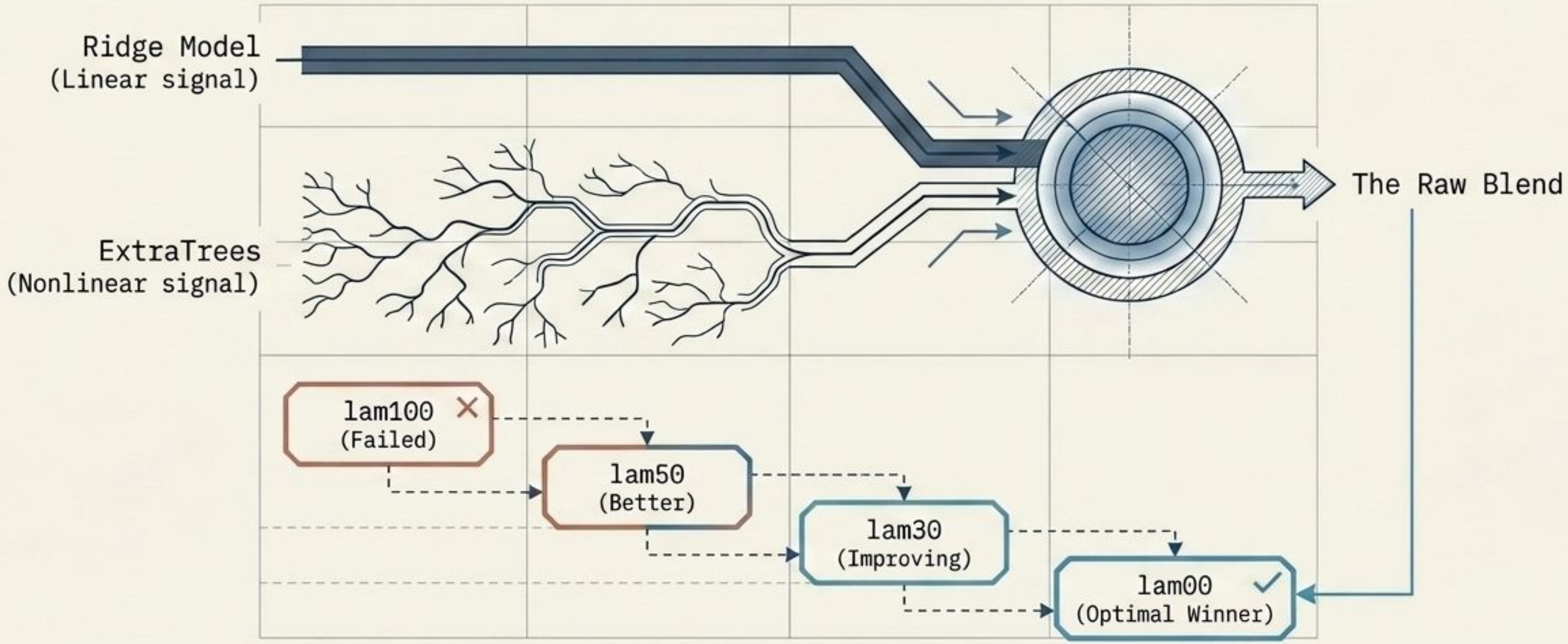
- Two-Stage Model

Note: Stage 1 Spline (Id) mapped to Stage 2 Ridge (Residuals). Local CV improved to 0.0241 but it catastrophically failed to generalize on Kaggle.



Expanding features or deploying standalone complex models failed. A fundamentally new architecture was required.

The Breakthrough Strategy: Raw Ensembling



My best algorithm abandoned single-model dominance and forced calibration. A raw probability blend of shallow trees and heavy linear regularization proved vastly superior.

The Engine Mechanics: under the hood

Ridge Regression

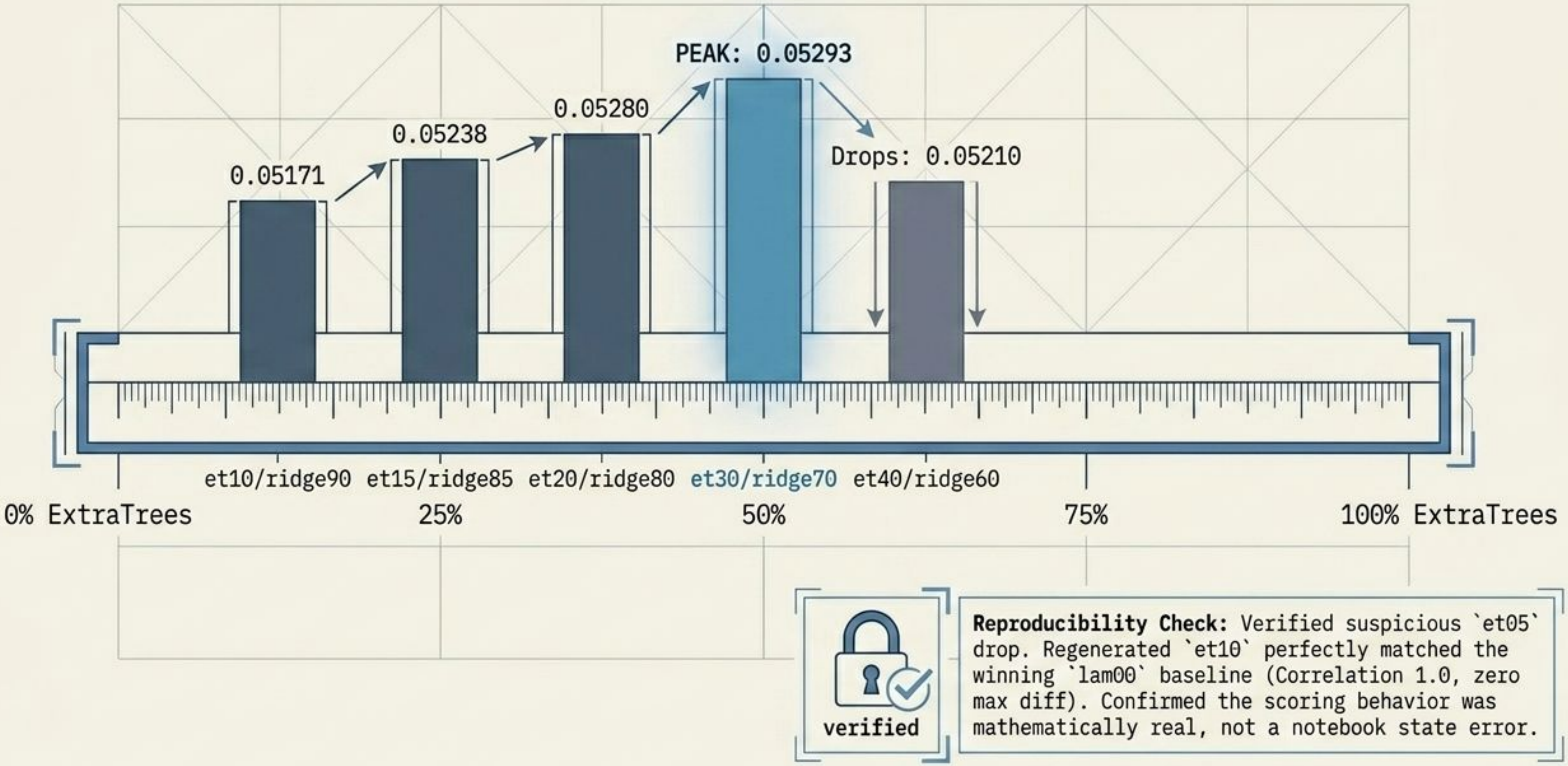
- **Signal Captured:** Broad, stable linear trends.
- **Regularization:** Highly penalized (alpha=3000) to prevent weight explosion on noisy features.
- **Role:** The reliable anchor.

ExtraTrees

- **Signal Captured:** Weak, highly localized nonlinear noise.
- **Regularization:** Kept deliberately shallow to avoid memorizing noise.
- **Role:** The nonlinear signal detector.

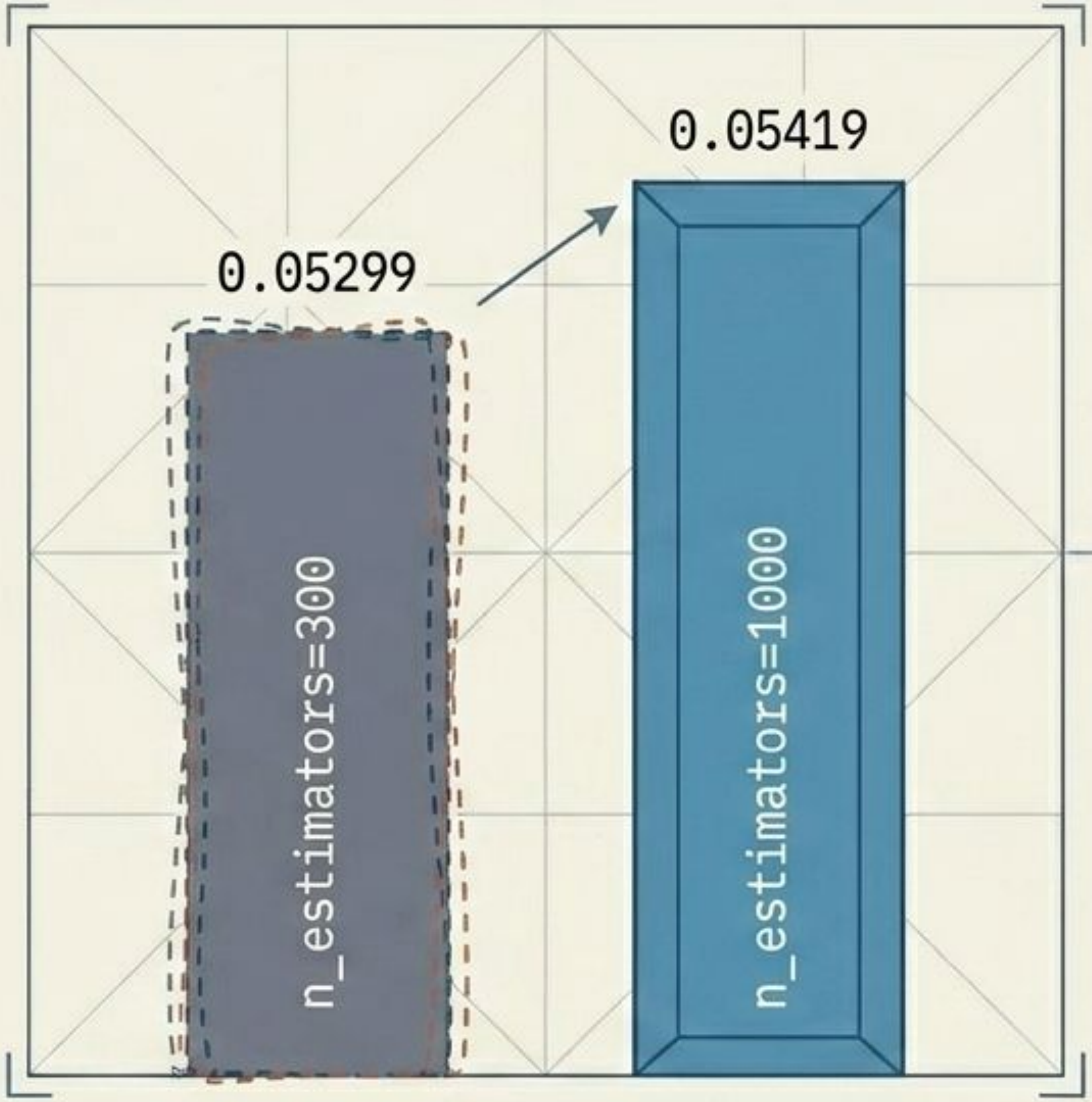
They cover each other's blind spots. Ridge provides the macro-structure stability, while ExtraTrees captures the micro-variations.

Finding the Sweet Spot



Reproducibility Check: Verified suspicious `et05` drop. Regenerated `et10` perfectly matched the winning `lam00` baseline (Correlation 1.0, zero max diff). Confirmed the scoring behavior was mathematically real, not a notebook state error.

Stabilizing the Nonlinear Signal

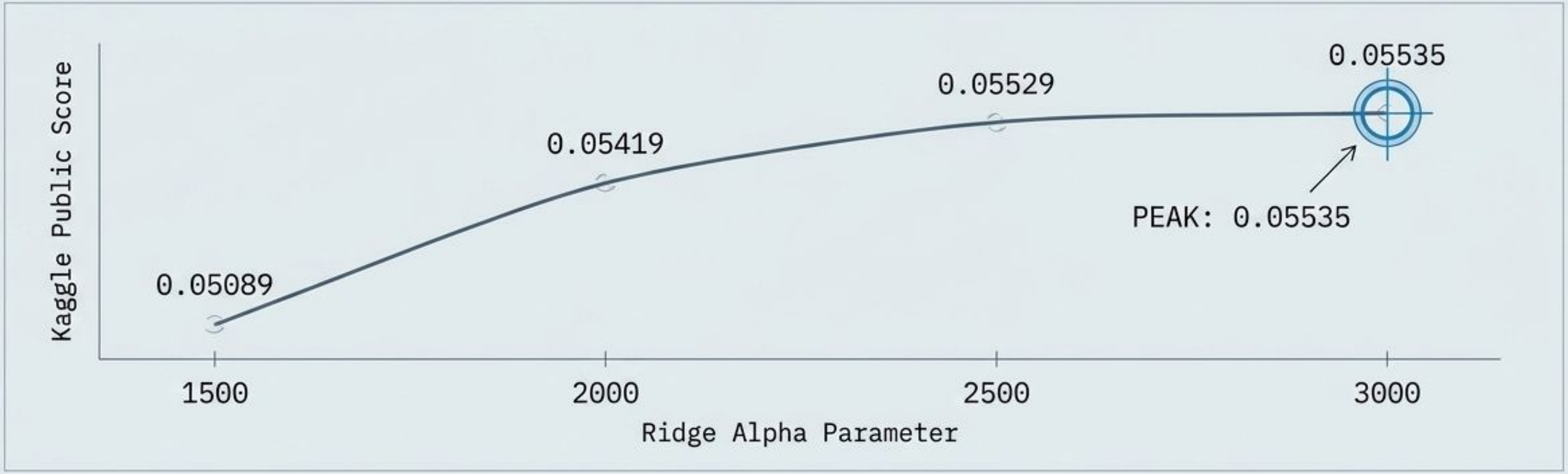


(Based on the et25/ridge75 blend architecture)

The nonlinear signal captured by ExtraTrees was real but intrinsically noisy.

Massively increasing the estimator count provided necessary averaging stability, smoothing variance without adding model complexity.

Constraining the Weights



Theory Alignment
Validating the Lecture: Ridge introduces L2 regularization. Pushing the penalty up to 3000 actively prevented the model from overly relying on noisy features, dramatically improving generalization.

The Danger of Depth

Baseline Score

Actual Score: 0.05535

Test 1: Increase Depth

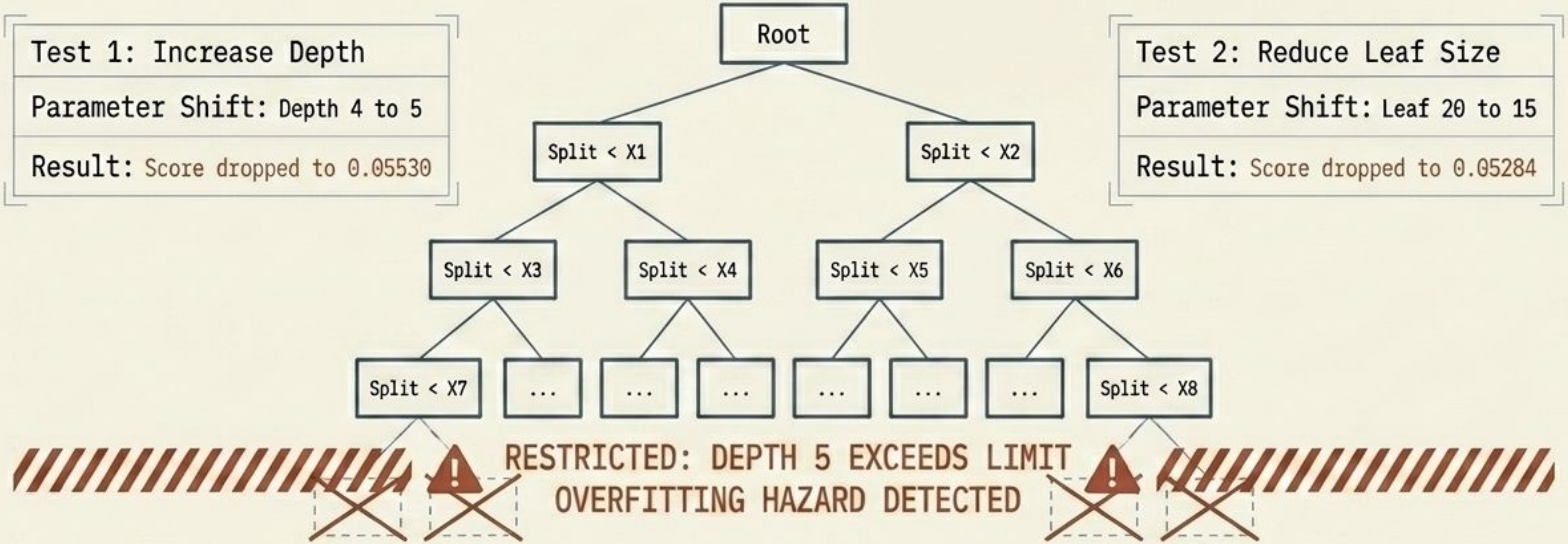
Parameter Shift: Depth 4 to 5

Result: Score dropped to 0.05530

Test 2: Reduce Leaf Size

Parameter Shift: Leaf 20 to 15

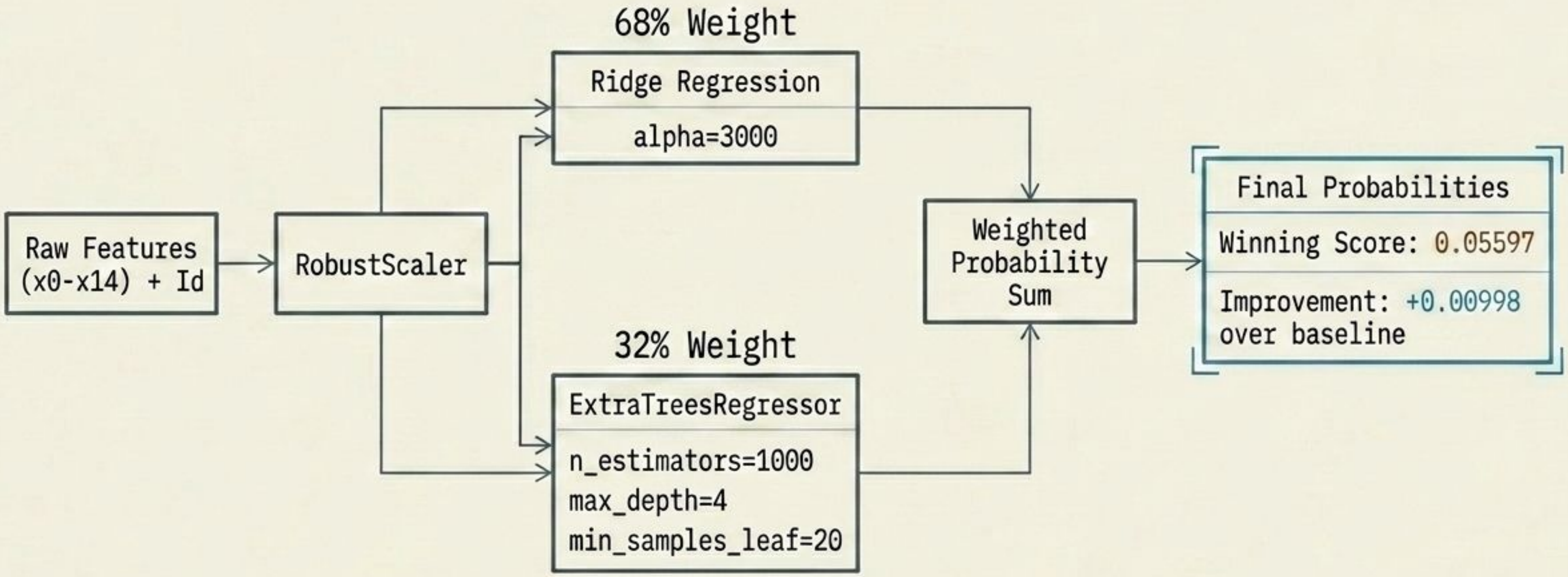
Result: Score dropped to 0.05284



Final Insight

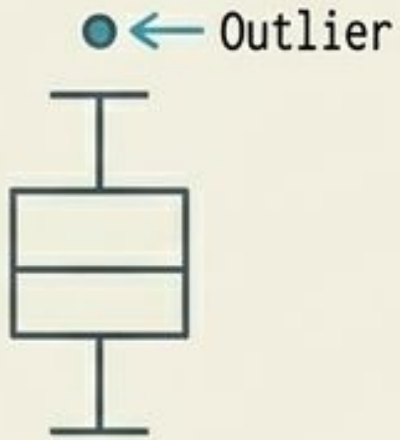
Final complexity checks confirmed that ExtraTrees performed best as a highly constrained, shallow ensemble. Deeper trees immediately memorized training noise and destroyed generalization capability.

My Best Architecture Blueprint



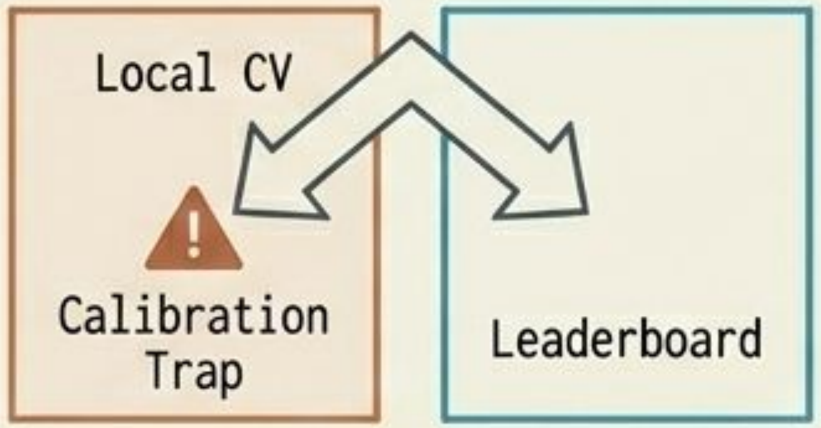
Post-Mortem & Theoretical Alignment

Signal in the Extremes



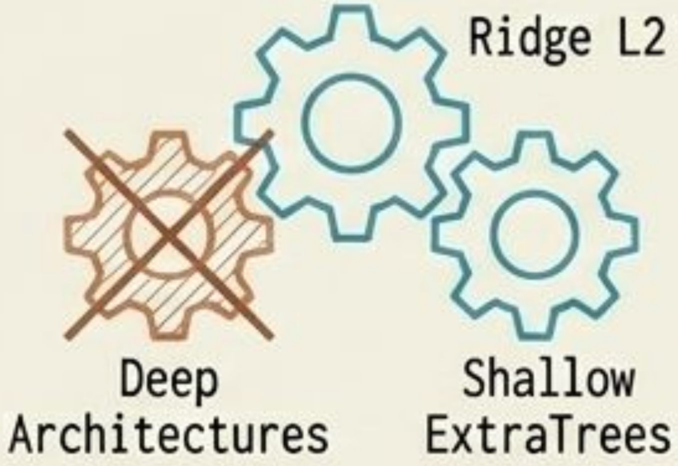
Outliers were structural, not errors. Robust scaling preserved the signal while preventing gradient disruption, proving mathematically superior to brute-force clipping.

The Limits of Local Validation



Models optimizing squared error (R^2) are highly sensitive to sample variance. Balancing local CV with Leaderboard reality was absolutely essential to avoid the calibration trap.

Ensembling over Complexity



Blending a stable linear regularizer (Ridge L2) with a weak nonlinear detector (Shallow ExtraTrees) easily outperformed standalone complex deep architectures.

Code:

https://github.com/later8/ml-hw-spring-2026-python/blob/29f9fe100af55bc8aa0e65fd58c7d07206df92a4/kaggle_final.ipynb